
Timed reference
y
x
r
v_d
$x_d,\ v_d,\ a_d,\ \psi_d$
Geometric feedback
$e_x = x - x_d, \quad e_v = v - v_d$
Position and attitude errors
$b_{3d} = A/\|A\|$
$f = A^{\mathrm{T}} R e_3, \quad M \in \mathbb{R}^3$
Rotor allocation
$u = F_a^{-1} \begin{bmatrix} f \\ M \end{bmatrix}$
$\omega_i = \mathrm{sgn}(u_i)\sqrt{|u_i|}$
Four signed rotor commands
Quadrotor dynamics
$x,\ v,\ R,\ \Omega$
State feedback: position, velocity, attitude and body rate